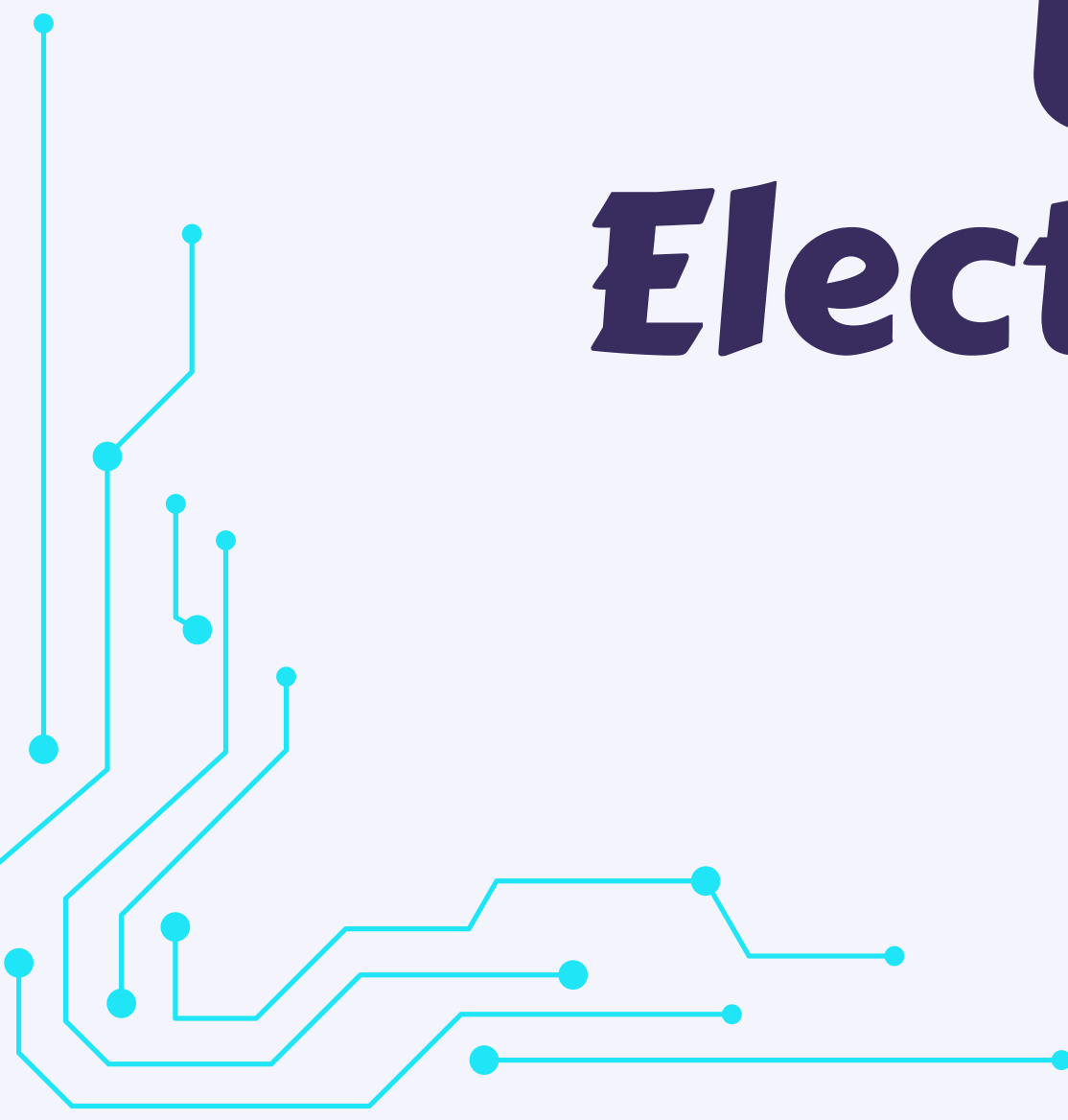
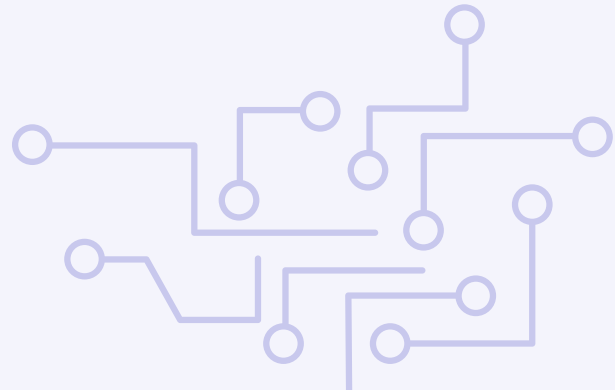
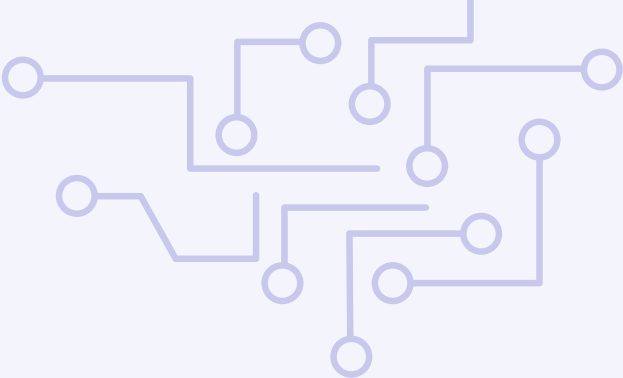
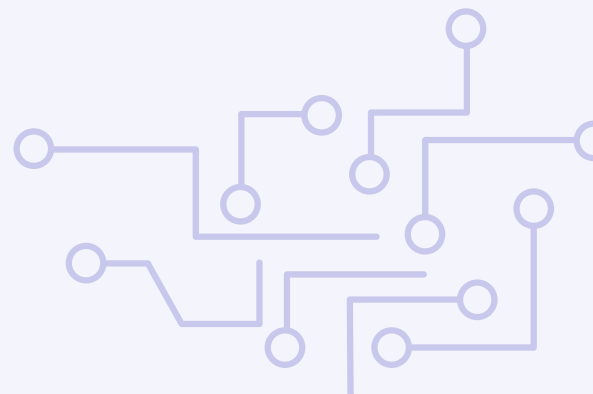
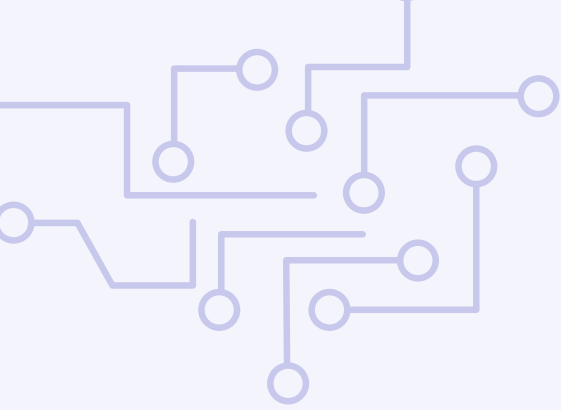
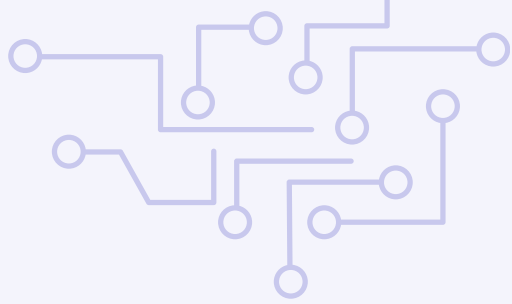




Soldering Circuits & Understanding Electrical Components

Attendance:



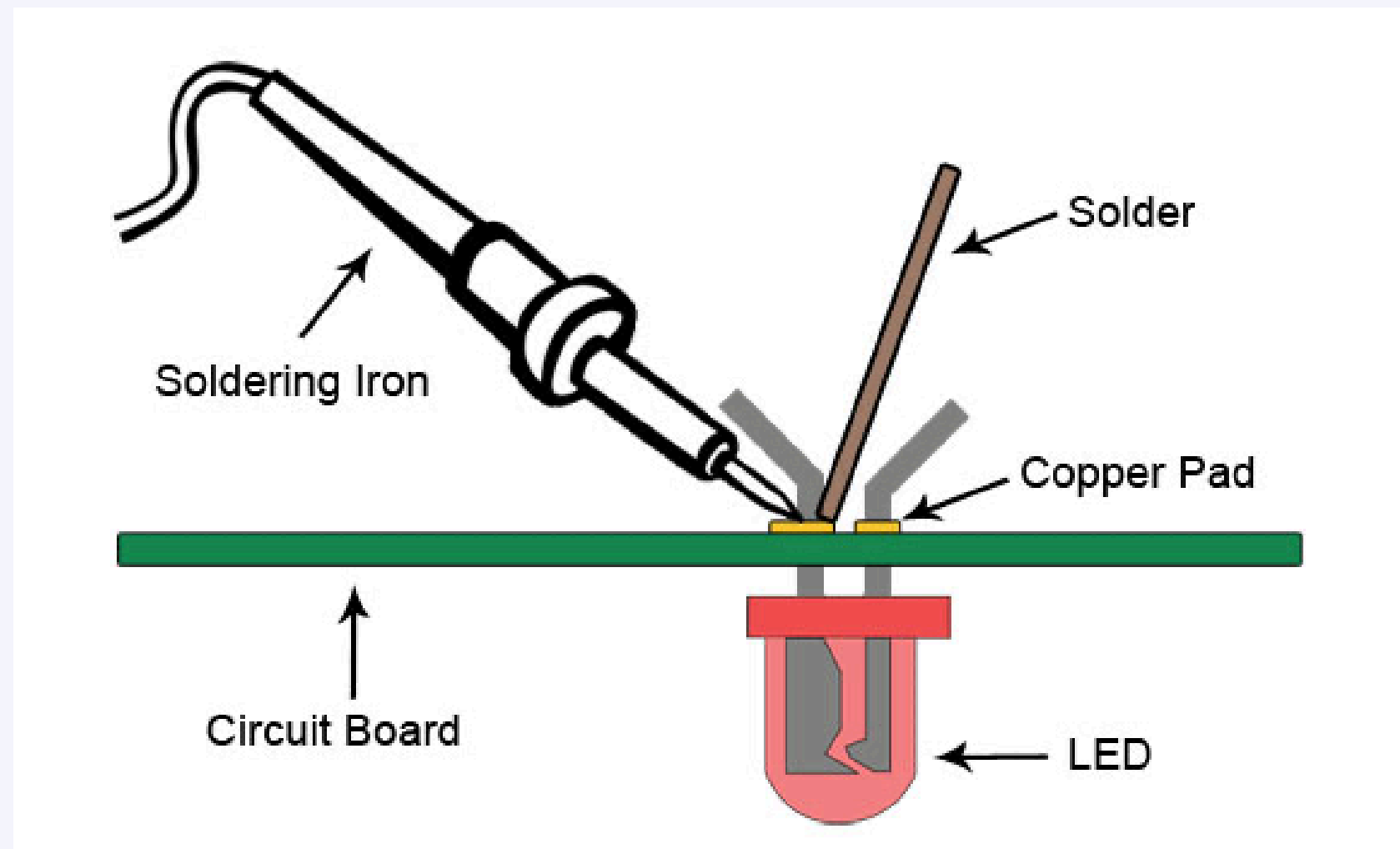


Soldering

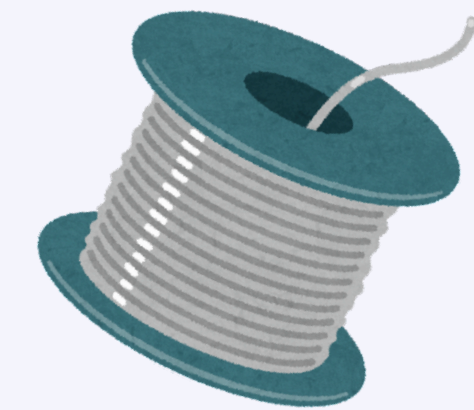


SOLDERING IS THE PROCESS OF CONNECTING METAL COMPONENTS TOGETHER BY MELTING A FILLER METAL, CALLED SOLDER, USING HEAT. AS THE METAL COOLS AND SOLIDIFIES, IT CREATES A STRONG, DURABLE, AND CONDUCTIVE BOND, ALLOWING ELECTRICITY TO FLOW BETWEEN.

**IT CAN BURN YOU!
BE CAREFUL!**



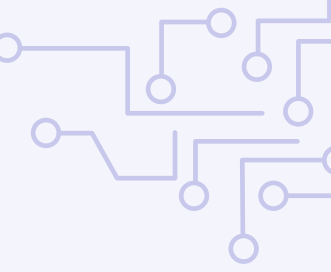
(Filler Metal - Solder)



make it look like a Kisses



Components of the Circuit



- **Resistors**



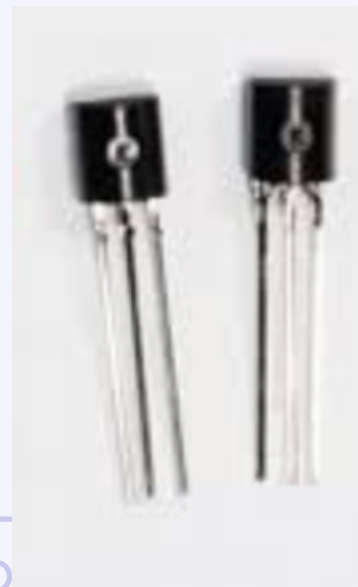
- **Vibration Switch**



- **104 Ceramic Capacitors**



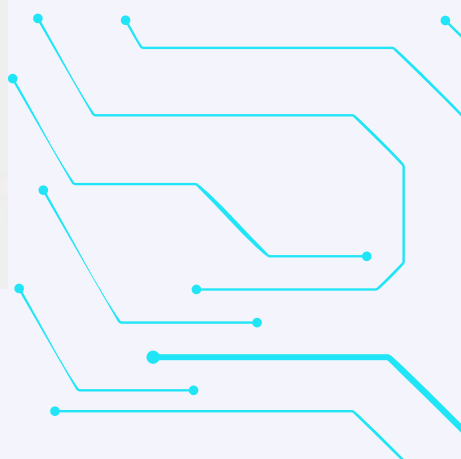
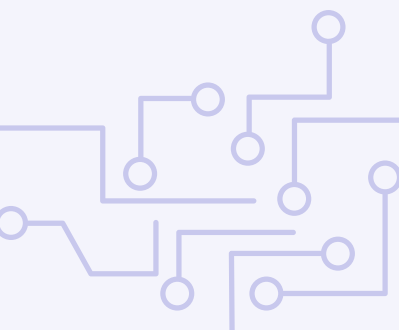
- **9013 Transistors**

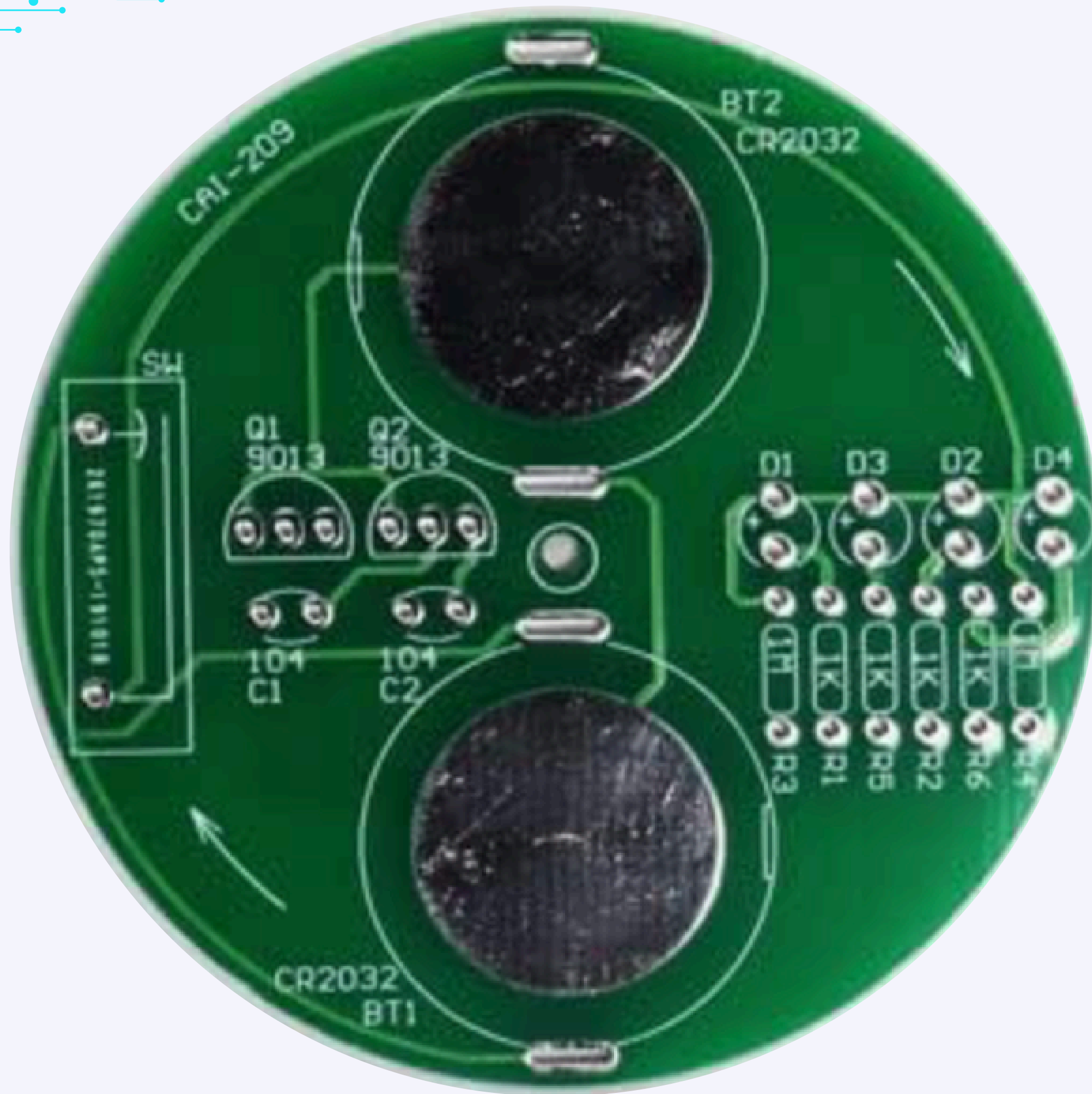


- **Button Battery Clips**



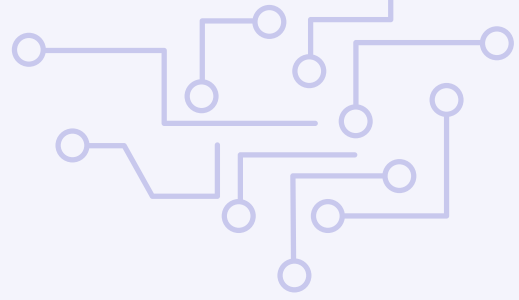
- **LED Diodes**





- ***R = Resistor***
- ***D = LED Diode***
- ***BT = Battery Clip***
- ***Q = Transistor***
- ***SW = Switch***
- ***C = Capacitor***
- ***Q = Triode***

Resistors



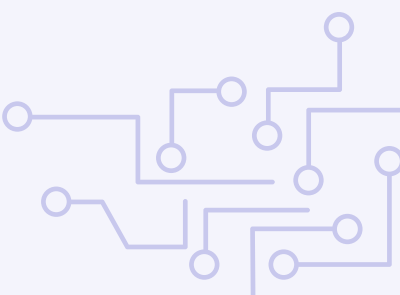
$R1, R2, R5, \text{ and } R6 = 1\text{k}\Omega = 1,000\Omega$ | $R3 \text{ and } R4 = 1\text{M}\Omega = 1,000,000\Omega$

What is it?

- Two-terminal electrical component
- Limits the amount of current
- Measured in Ohms (Ω)

Function on Circuit

- Manage the voltage level to protect the LEDs from receiving too much current



Capacitors

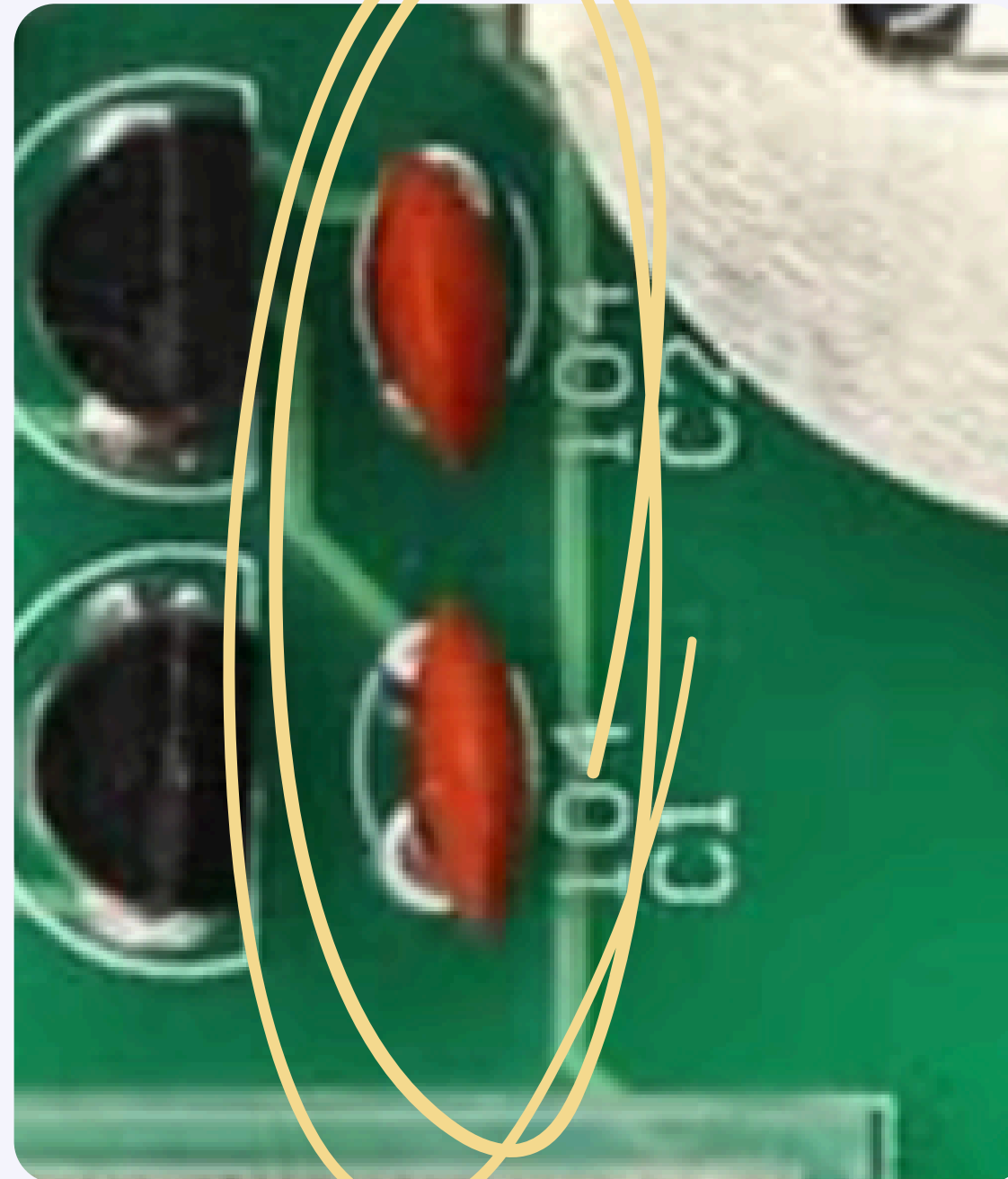
$$C1, C1 = 0.1\mu F = 0.00000001$$

What is it?

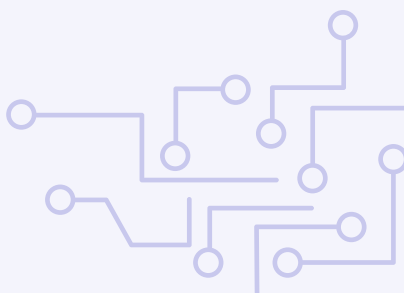
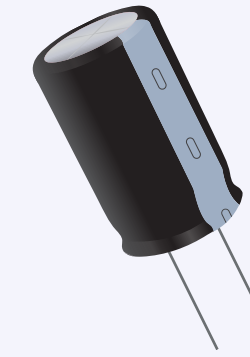
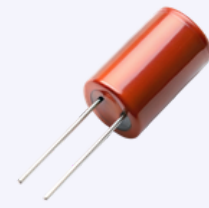
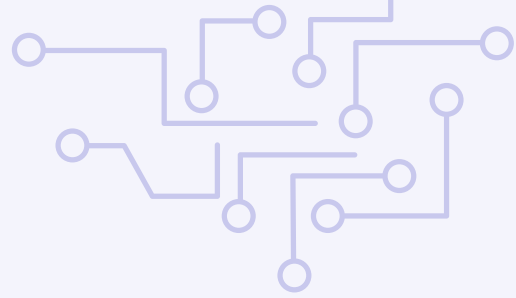
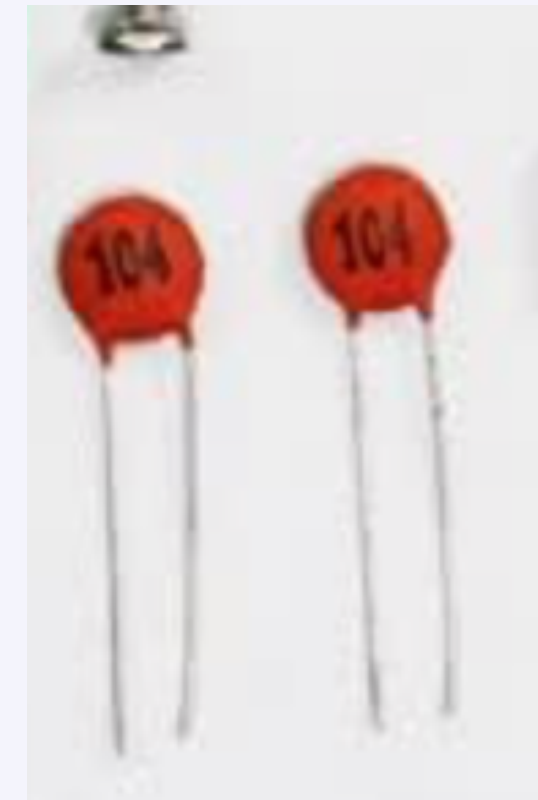
- Energy reserves
- Reduce voltage spikes and filter out noise
- Dielectric or non-conductive layer that keeps internal metal connections apart
- Measured in Farads (F)

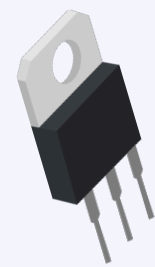
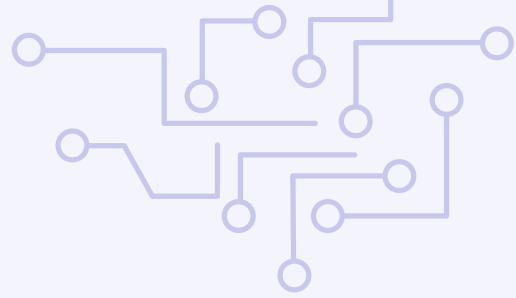
Function on Circuit

- The time it takes for a capacitor to store a small amount of electrical charge and then release it determines how fast the LEDs blink



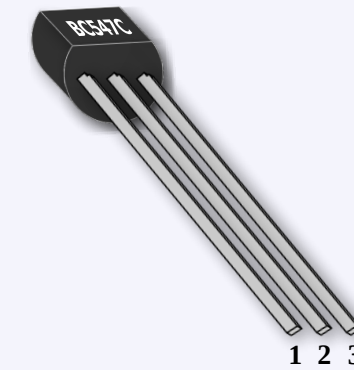
0.1 μ F





9013 Transistors

Q1, Q2

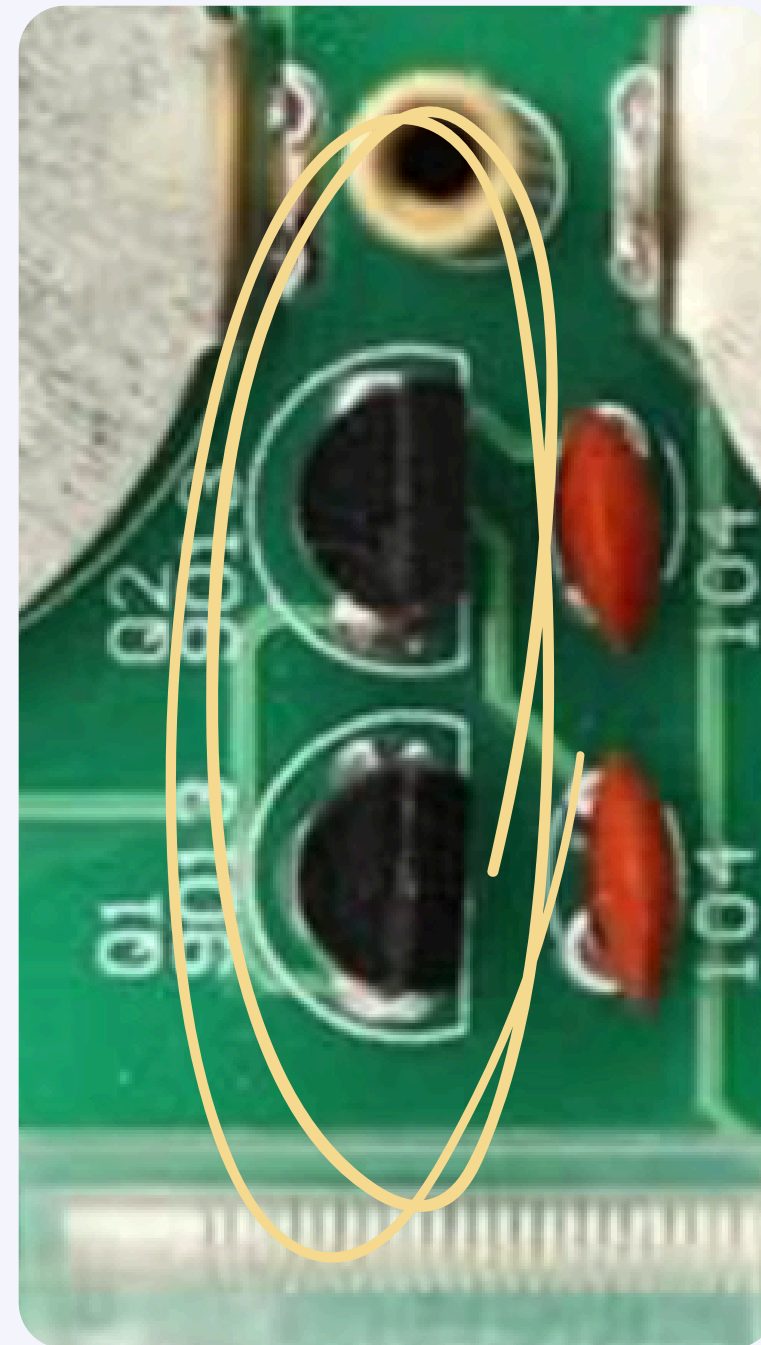


What is it?

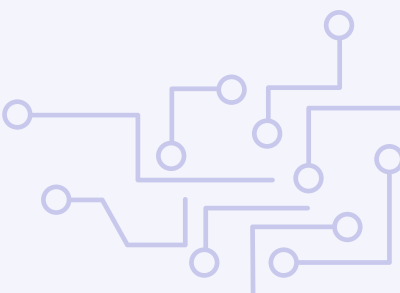
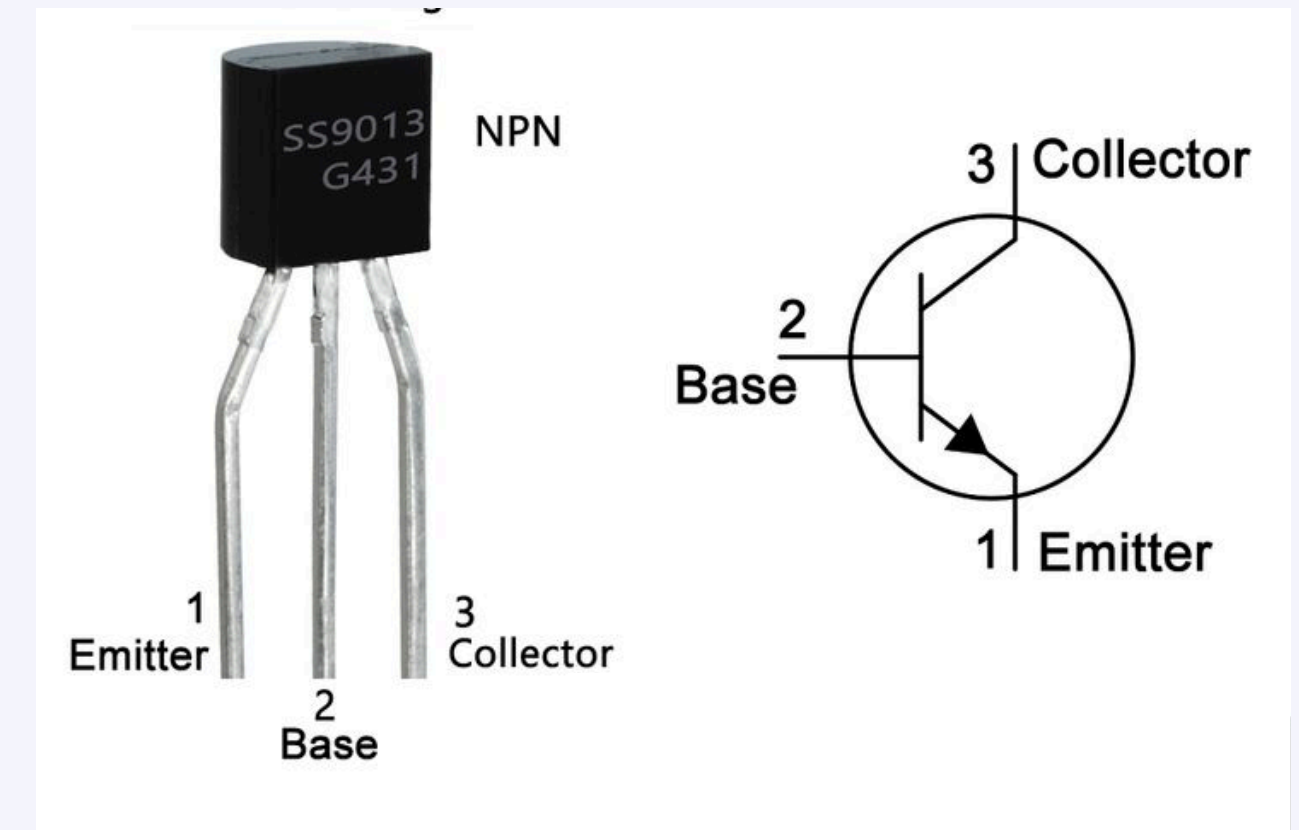
- Negative-Positive-Negative (NPN) transistor
- Serves as an electrical switch, turning current on/off

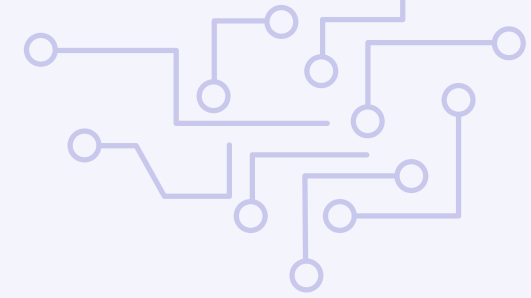
Function on Circuit

- "open" and "close" to allow electricity to reach the LEDs
- When Q1 is "on," one set of LEDs lights up; when Q1 turns "off," Q2 turns "on"

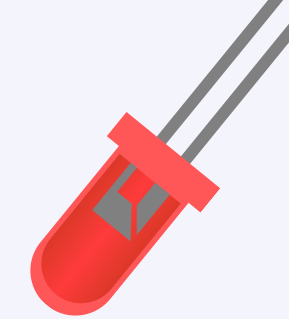


- Left Pin: Emitter
- Center Pin: Base
- Right Pin: Collector





LED Diodes



D1, D2, D3, D4

What is it?

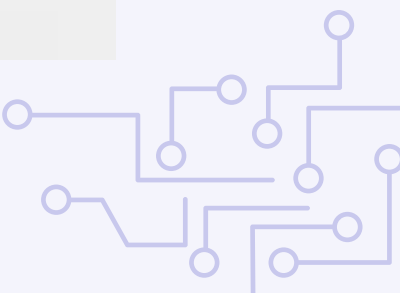
- A two-terminal semiconductor
- Emits light
- One-way valve
- Has polarity

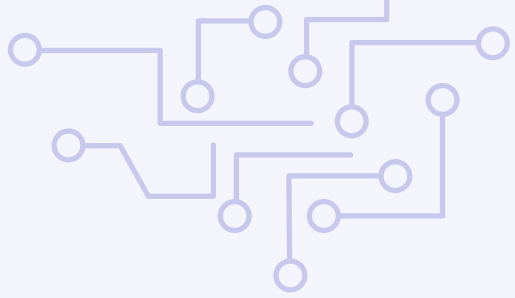
Function on Circuit

- Visual status: Helps let you know if the circuit works!



Anode - Longer side (+)
Cathode - Shorter side (-)





Vibration Switch



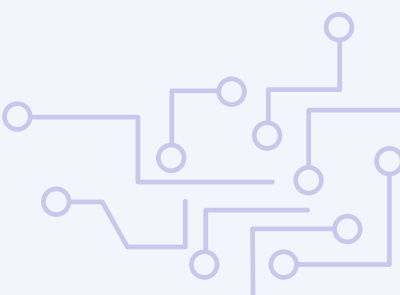
What is it?

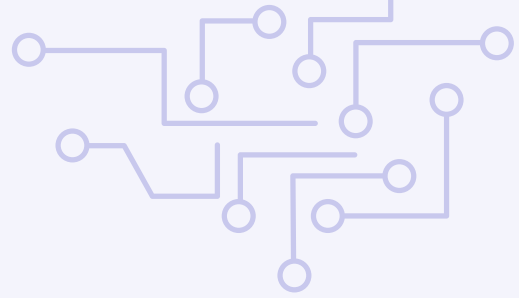
- An electronic system that detects mechanical movement
- Acts as a sensor to trigger an action
- Completes or breaks an electrical connection

SW1

Function on Circuit

- Senses the spinning motion and turns the LEDs on





Button Battery Clips



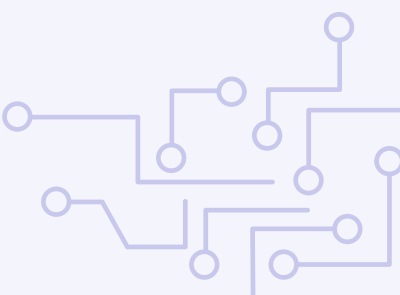
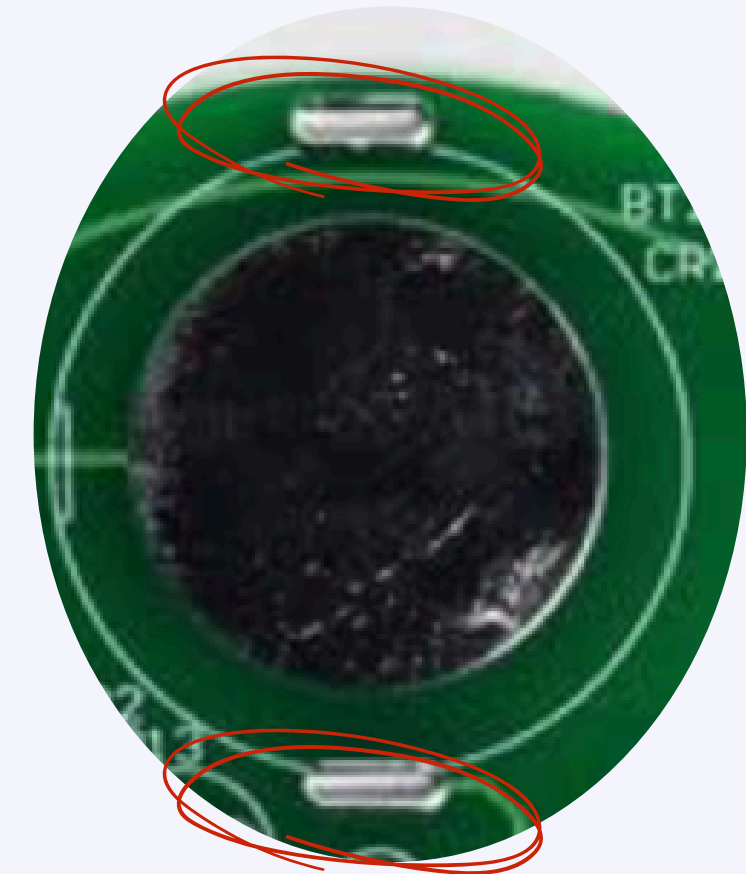
What is it?

- Conductive, metal components designed to physically hold coin-cell batteries
- Ensure consistent pressure on the battery terminals
- Facilitate power transfer

Function on Circuit

- Connect the battery poles to the circuit, enabling current flow

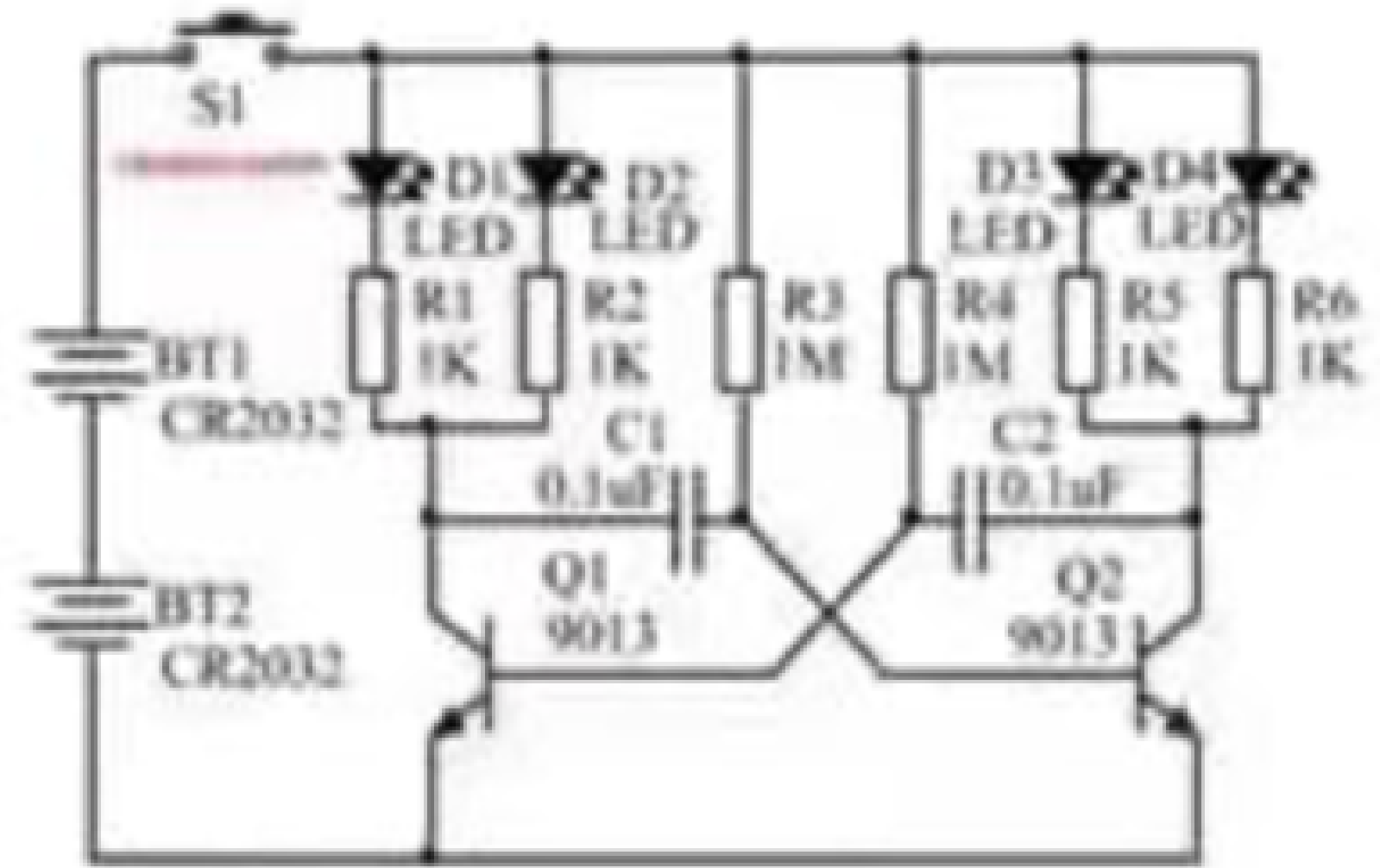
BT1, BT2

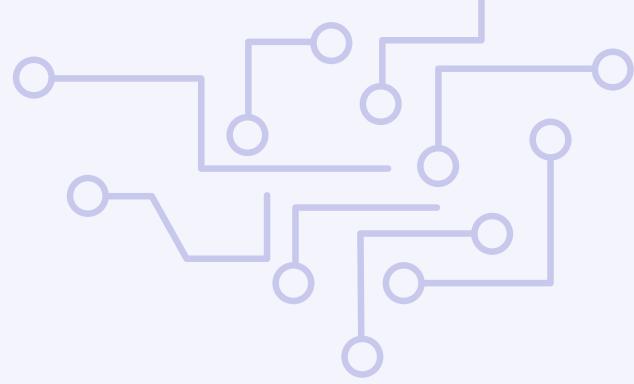
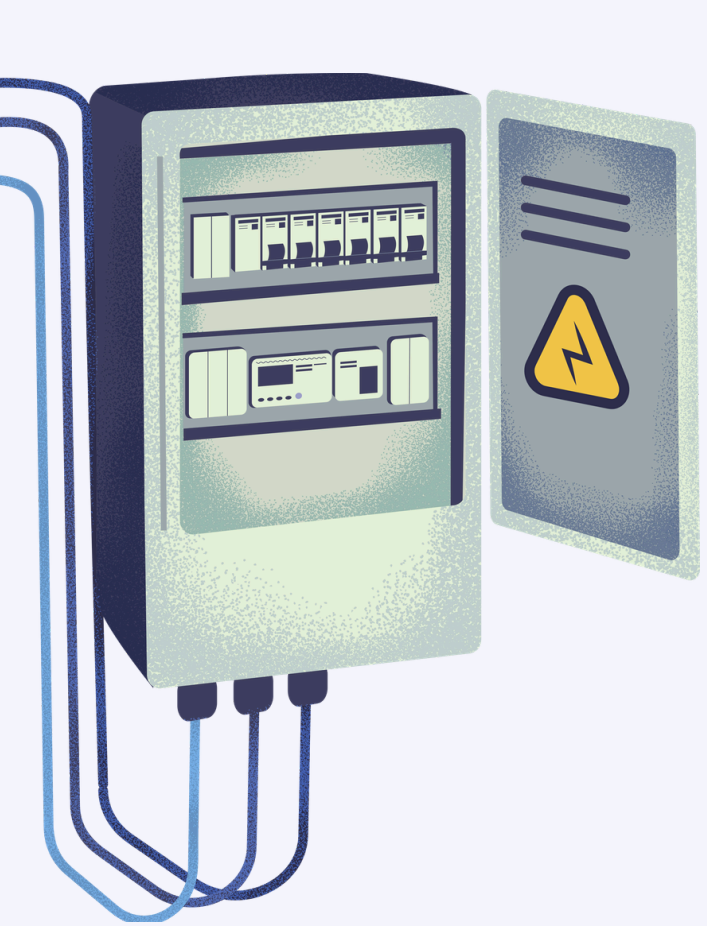


FINISHED!



Circuit schematic





***Thank you for
coming! <3***

